

Sakthivel G

Senior Unity Developer | XR Simulation Engineer

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PROFESSIONAL SUMMARY

Senior Unity Developer with 5+ years of experience designing and developing enterprise-grade VR training applications, industrial simulations, AR solutions, and cross-platform interactive experiences using Unity. Specialized in Meta XR SDK, C#, WebGL optimization, IoT hardware integration (ESP32), and scalable software architecture for real-time applications. Experienced in building immersive training solutions for Meta Quest, PC, mobile, and WebGL platforms, with expertise in simulation systems, performance optimization, REST API integration, multilingual localization, and modular development. Passionate about delivering high-quality, production-ready applications, mentoring junior developers, and continuously learning and adopting emerging real-time technologies such as Unreal Engine and real-time 3D workflows.

PROFESSIONAL EXPERIENCE

VyVoxel 🌐	2021 – 2022
<i>Junior Unity Developer</i>	Chennai
• Developed AR and VR simulation applications using Unity. • Implemented UI systems, gameplay mechanics, and interactive training workflows. • Created reusable, modular, and scalable code architecture for maintainable projects. • Integrated animations, particle effects, audio, video, and other media components. • Optimized scenes using Unity Light Baking for improved visual quality and performance. • Used Git and GitHub for version control, collaboration, and source code management. • Gained hands-on experience with Unreal Engine, exploring real-time rendering, Blueprints, and VR development workflows.	
Energetics AI 🌐	05/2022 – 08/2024
<i>Senior Technical Artist</i>	Chennai
• Developed cross-platform interactive applications and cognitive games using Unity, delivering optimized experiences across Android, iOS, PC, and WebGL. • Optimized 3D assets, materials, shaders, lighting, and rendering for high-performance real-time applications. • Implemented gameplay systems, UI, animations, particle effects, audio, and media integration.	

- Collaborated with artists, designers, and developers to transform creative concepts into polished, production-ready experiences.
- Created reusable and scalable C# architecture to improve development efficiency and maintainability.
- Performed scene optimization, light baking, and performance tuning to ensure smooth gameplay across multiple platforms.
- Used **Git/Bitbucket** for version control, collaboration, and project management.
- Gained hands-on experience with **Unreal Engine** and real-time content creation workflows.

iRex Studio

09/2024 – 08/2025
Chennai

Senior Unity Developer

- Led the development of Unity-based **VR training simulations, product visualization, and real estate visualization** applications using C#.
- Developed immersive VR experiences using **Meta XR SDK, OpenXR, and AR Foundation**.
- Collaborated directly with clients to gather requirements, propose technical solutions, and deliver features aligned with business objectives.
- Designed and implemented scalable gameplay systems, simulation mechanics, UI workflows, and reusable architecture.
- Integrated REST APIs, IoT devices, and third-party SDKs to enhance application functionality.
- Optimized application performance through profiling, light baking, asset optimization, and rendering improvements for Meta Quest, PC, and WebGL.
- Mentored junior developers, conducted code reviews, and collaborated with cross-functional teams to deliver production-ready solutions on schedule.

Spatio

09/2025 – Present
Chennai

Senior Unity Developer

- Developed immersive VR training applications, industrial simulations, product visualization, and real estate visualization solutions using Unity, C#, and Meta XR SDK.
- Built interactive training modules for Forklift Operations, EOT Crane, Mechanical Pump, Heat Straightening, PPE, Work at Height, Manual Handling, and Electrical Safety.
- Designed and implemented scalable gameplay systems, simulation mechanics, UI workflows, evaluation systems, checkpoints, and scoring logic.
- Integrated ESP32 IoT hardware, Thrustmaster steering wheel, REST APIs, JSON-based content management, multilingual localization, analytics, and LMS connectivity.
- Optimized applications for Meta Quest, PC, and WebGL through light baking, asset optimization, memory management, and rendering enhancements.
- Developed reusable C# architecture, editor tools, and modular systems while mentoring junior developers and promoting coding best practices.

- Collaborated with project managers, UI/UX designers, and 3D artists to implement requirements and deliver production-ready solutions within project timelines.
- Managed source control using Git/GitHub and followed Agile development practices throughout the software development lifecycle.

EDUCATION

Image Creative Education	12/2018 – 01/2021
<i>C-Tech Higher Diploma in Game Technologies</i>	Vadapalani, Chennai
Shanmuga Industries Arts & Science College	2015 – 2018
<i>B.A English literature</i>	Tiruvannamalai, Tamil Nadu

TECHNICAL SKILLS

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|----------------------------|------------------------|--------------------------------|
| • Unity 3D | • Unreal Engine | • C# |
| • Threejs | • WebGL | • VR Simulation |
| • AR Foundation | • Meta SDK | • XR Interaction Toolkit (XRI) |
| • IoT Integration (ESP32) | • REST API Integration | • Unity Light Baking |
| • Git & GitHub | • Maya | • UI Development |
| • Performance Optimization | • Figma | • Adobe Illustrator |

LANGUAGES

English	Tamil
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KEY PROJECTS

Industrial VR Training

- Forklift Operations Simulator
- EOT Crane Training Simulator
- Mechanical Pump Assembly & Disassembly
- Heat Straightening Training
- PPE Training
- Work at Height Training
- Manual Handling Training
- Electrical Safety Training

VR & Interactive Solutions

- Data Center VR Experience
- Product Visualization
- Real Estate Visualization

Interactive Applications

- Cognitive Games
- AR & VR Applications

SOFT SKILLS

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|----------------------|-----------------------|
| • Problem Solving | • Technical Mentoring |
| • Team Collaboration | • Time Management |
| • Communication | • Adaptability |
| • Documentation | • Continuous Learning |